

MovementLogger — Sensor & Module Catalog

11 Teile · 2026-05-21 11:10 UTC · BOM aus movement_logger_firmware · via crawl2pump

11 neu (7 Tage)

0 aktualisiert

· 101 touched · 0 price changes

Dev Boards 4

STEVAL-MKBOXPRO (SensorTile.box PRO Rev_C)

OSS firmware

USB-C pluggable

☒ USB-C

☐ WiFi

☒ Bluetooth

☐ GPS

☒ Motion sensors

☒ SD-card

L×B×H 6.3 × 4 × 2 cm · 94 g

MCU: STM32U585 · Cortex-M33 @160 MHz · 2 MB flash / 786 KB SRAM · TrustZone · ultra-low-power

Datasheet: https://www.st.com/resource/en/user_manual/um3133-sensortilebox-pro-stmicroelectronics.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



neu STEVAL-MKBOXPRO (SensorTile.box PRO Rev_C) · STEVAL-MKBOXPRO @ DigiKey

CHF 94.56

STMicroelectronics @ DigiKey · in stock

Primary target board. USB-C for charge + STM32 DFU flashing. · USB-C · firmware: open-source

<https://www.digikey.com/en/products/detail/stmicroelectronics/STEVAL-MKBOXPRO/19721662>

GPS / GNSS 22

AliExpress FC 14-pin Cable — 1.27 mm IDC → DuPont female, 30 cm (2X7P variant)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 30 × 0.5 × 0.3 cm · 7 g



neu AliExpress FC 14-pin Cable — 1.27 mm IDC → DuPont female, 30 cm (2X7P variant) · Vino Electronic Connect. (AliExpress) @ Vino Electronic Connect. (AliExpress) (vendor page)

Vino Electronic Connect. (AliExpress) (manufacturer store)

****PIN CHOICE IS CRITICAL** — wrong pins can brick SWD debug or destroy GPS **** Use ONLY 3 of the 14 wires:** · ****pin 13**** (UART4_RX on STEVAL) → GPS ****TX**** · ****pin 14**** (UART4_TX on STEVAL) → GPS ****RX**** · ****pin 7 or 11**** (GND) → GPS ****GND**** ALL OTHER PINS carry SWD/JTAG signals — ****do NOT touch**** (bridging any of them can permanently lock out debug access or damage the GPS). Identify pin 1 by the red/blue stripe on the ribbon (standard IDC marking — pin 1 is on the same side as the marked wire). The header is 2-row × 7-col: odd pins (1,3,5,7,9,11,13) on one row, even pins (2,4,6,8,10,12,14) on the other; pin 13/14 are the LAST pair at the far end of the connector. If unsure, multimeter continuity test: STEVAL off, probe JP2 pin 1 silkscreen marker against each wire end. GPS VCC (3.3 V) comes from JP4, NOT JP2 — use the Samtec SSW-107-01-G-S header card below + one F/F DuPont from the...

<https://de.aliexpress.com/item/1005011846708876.html>

GPS/GNSS Magnetic Mount Antenna 3 m (active, SMA)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 4 × 4 × 1.3 cm · 80 g

Datasheet: <https://www.sparkfun.com/products/14986>



neu **GPS/GNSS Magnetic Mount Antenna 3 m (active, SMA) · GPS-14986 @ DigiKey**

SparkFun @ DigiKey · out of stock

Active GPS antenna (~28 dB LNA, 3 m RG-174 coax, SMA male). Magnetic base — sits on the board's non-metal top or the case lid for best sky view. Needs a U.FL → SMA pigtail to mate with the MAX-M10S breakout's U.FL connector — see next part. · SMA / U.FL · firmware: open-source

<https://www.digikey.com/en/products/detail/sparkfun-electronics/14986/9682235>

CHF 12.85

Interface Cable SMA → U.FL, 100 mm (pigtail)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 10 × 0.3 × 0.3 cm · 3 g

Datasheet: <https://www.sparkfun.com/products/9145>



neu **Interface Cable SMA → U.FL, 100 mm (pigtail) · SparkFun 9145 @ sparkfun.com**

SparkFun

100 mm pigtail: U.FL female (plugs into the SparkFun MAX-M10S breakout) ↔ SMA female bulkhead (mates with the antenna above's SMA male). Buy with the magnetic antenna or skip both if the onboard chip antenna is enough. · SMA / U.FL · firmware: open-source

<https://www.sparkfun.com/products/9145>

USD 5.75

Samtec FFSD-07-D-04.00-01-N — 14-pin 1.27 mm IDC ribbon (100 mm)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

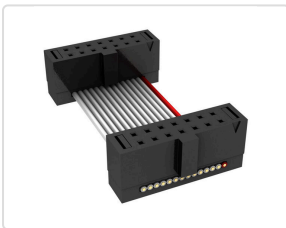
☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 10 × 0.5 × 0.2 cm · 3 g

Datasheet: https://suddendocs.samtec.com/catalog_english/ffsd.pdf



neu **Samtec FFSD-07-D-04.00-01-N — 14-pin 1.27 mm IDC ribbon (100 mm) · FFSD-07-D-04.00-01-N @ DigiKey**

Samtec @ DigiKey · in stock

Solderless JP2 plug (the FTSH-107 programming header). 1.27 mm shrouded sockets on BOTH ends, so to actually reach the SparkFun MAX-M10S breakout you build a 4-SKU assembly: (1) **this cable** (DigiKey CHF 9.21, real photo) for the JP2 plug; (2) a **1.27 → 2.54 mm SWD adapter PCB** (~CHF 3 on AliExpress, no canonical Mouser/DigiKey MPN — the entire Western-distributor catalog was surveyed and this category is generic OEM only); (3) **4× female-female DuPont jumpers** for the adapter PCB → GPS breakout pins (Bastelgarage ~CHF 4 / 10 Stk); (4) **2.54 mm 7-pin female header** onto JP4 for the 3.3 V tap (~CHF 1, meter 3.25–3.40 V before connecting!). Pin map: JP2 pin 13 → GPS TX, pin 14 → GPS RX, pin 7 or 11 → GPS GND (crossover: STEVAL TX → GPS RX). Single AliExpress search term that surfaces this category end-to-end: "14 Pin 1.27mm SWD to Dupont Cable" — typically CHF 5, seller volatile....

<https://www.digikey.com/en/products/detail/samtec-inc/FFSD-07-D-04-00-01-N/6613350>

CHF 9.21

Samtec SSW-107-01-G-S — 1×7 female header strip, 0.1" pitch (JP4 power tap)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

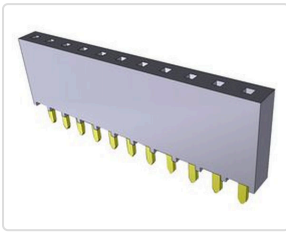
☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 1.8 × 0.85 × 0.25 cm · 0.50 g



neu Samtec SSW-107-01-G-S — 1×7 female header strip, 0.1" pitch (JP4 power tap) · SSW-107-01-G-S @ Mouser

CHF 1.34

Samtec @ Mouser · in stock

1×7 gold-plated through-hole socket strip on 2.54 mm (0.1") pitch. Slips onto JP4, the STEVAL's 7-pin power header — no soldering, friction-fit. Before plugging anything in: **set the JP4 domain switch to 3 V** (not 1.8 V) and meter the 3 V pin under power; must read 3.25–3.40 V. Then DuPont-jumper the 3 V pin to the SparkFun MAX-M10S breakout's VCC pin. The strip's other 6 pins stay accessible for GND tapping or for future signal-monitoring. · SMA / U.FL · firmware: open-source

<https://www.mouser.ch/ProductDetail/Samtec/SSW-107-01-G-S?qs=rU5faygh%252BE2WDJ3AF5mWdA%3D%3D>

SparkFun GPS Breakout - MAX-M10S (Qwiic)

OSS firmware

Qwiic / I²C

☐ USB-C

☐ WiFi

☐ Bluetooth

☒ GPS

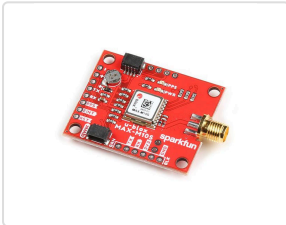
☐ Motion sensors

☐ SD-card

L×B×H 2.54 × 2.54 × 0.6 cm · 4 g

Datasheet: https://cdn.sparkfun.com/assets/7/5/9/a/a/MAX-M10S_DataSheet_UBX-20035208.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



neu SparkFun GPS Breakout - MAX-M10S (Qwiic) · GPS-18037 @ DigiKey

CHF 35.80

SparkFun @ DigiKey · in stock

Recommended MAX-M10S carrier. Open-hardware breakout. Has an onboard chip antenna that works through the polycarbonate case, plus a U.FL connector for an external active antenna (see the two antenna parts below) if the signal needs a boost. · Qwiic / I²C · firmware: open-source

<https://www.digikey.com/en/products/detail/sparkfun-electronics/18037/16719314>

SparkFun PRT-12796 — F/F DuPont jumper wires, 6 in (20 Pack)

OSS firmware

SMA / U.FL

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 15 × 4 × 0.3 cm · 25 g



neu SparkFun PRT-12796 — F/F DuPont jumper wires, 6 in (20 Pack) · PRT-12796 @ Mouser

CHF 2.19

SparkFun @ Mouser · in stock

Female-to-female DuPont jumper wires, 6 in (~150 mm), 20-pack. Used in the solderless GPS bring-up for: (a) 3× links between the 1.27→2.54 mm SWD adapter PCB's 2.54 mm side and the SparkFun MAX-M10S breakout's UART header (TX/RX/GND); (b) 1× link from the JP4 7-pin female header's 3.3 V pin to the GPS VCC pin. You need 4 jumpers from the pack; the rest stay in your bin for the next project. Also useful for any future I²C / Qwiic-jumper experiments on the LilyGO build. · SMA / U.FL · firmware: open-source

<https://www.mouser.ch/ProductDetail/SparkFun/PRT-12796?qs=WYAARYrbSnYazPSZHWXCfQ%3D%3D>

u-blox MAX-M10S GNSS module

OSS firmware

UART pins

☐ USB-C

☐ WiFi

☐ Bluetooth

☒ GPS

☐ Motion sensors

☐ SD-card

L×B×H 1.01 × 0.97 × 0.25 cm · 0.60 g

Datasheet: https://content.u-blox.com/sites/default/files/MAX-M10S_DataSheet_UBX-20035208.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



neu u-blox MAX-M10S GNSS module · MAX-M10S-00B @ DigiKey

u-blox @ DigiKey · in stock

Wired to UART4 @ 38400. The firmware's GPS source. · UART pins · firmware: open-source

<https://www.digikey.com/en/products/detail/u-blox/MAX-M10S-00B/15712906>

CHF 10.38

Enclosure 7

Hammond 1554G2GYCL — IP66 PC enclosure, clear lid (120×90×60 mm)

OSS firmware

enclosure

☐ USB-C

☐ WiFi

☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 12 × 9 × 6 cm · 154 g

Datasheet: <https://www.hammfg.com/electronics/small-case/plastic/1554.pdf>



neu Hammond 1554G2GYCL — IP66 PC enclosure, clear lid (120×90×60 mm) · 1554G2GYCL @ DigiKey

Hammond Manufacturing @ DigiKey · in stock

Polycarbonate IP66 project box, external 120 × 90 × 60 mm, clear PC lid. Lid sealed with O-ring gasket and four corner screws. RF-transparent so onboard GPS (and any radar) reads through the wall. The Hall sensor reads a passing magnet through the lid, so you can power-cycle the recorder without opening the box. Leave the wall un-drilled to keep the seal intact and charge via Qi wireless through the wall (the STEVAL supports it). If the LPS22DF barometer needs ambient pressure, add a Gore-type vent. IP66 — splash and water-jet rated, NOT submersible. If the box is mounted on a foilboard that will wipeout-submerge, prefer the SERPAC RBF63 (IP67) below. · enclosure · firmware: open-source

<https://www.digikey.com/en/products/detail/hammond-manufacturing/1554G2GYCL/2359944>

CHF 22.77

SERPAC RBF53P06C10C — IP67 PC enclosure, clear lid (120 × 80 × 40 mm)

OSS firmware

enclosure

☐ USB-C

☐ WiFi

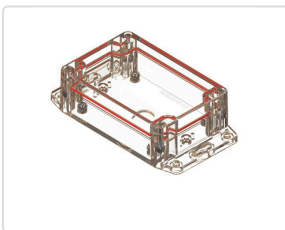
☐ Bluetooth

☐ GPS

☐ Motion sensors

☐ SD-card

L×B×H 12 × 8 × 4 cm · 75 g



neu SERPAC RBF53P06C10C — IP67 PC enclosure, clear lid (120 × 80 × 40 mm) · RBF53P06C10C @ DigiKey

SERPAC @ DigiKey · in stock

Polycarbonate IP67 project box, external 120 × 80 × 40 mm (4.72 × 3.15 × 1.59 in), clear PC lid sealed with perimeter O-ring + stainless-steel screws into metal inserts. IP67 = submersion to 1 m for 30 min, the right call for a foilboard recorder that will go under during wipeouts. Smaller footprint than the RBF63 line — the MovementLogger (STEVAL 63 × 40 mm + SparkFun MAX-M10S breakout 25 × 25 mm side-by-side ≈ 88 × 40 mm) fits cleanly inside the ~108 × 65 mm interior with no excess case length to flop around. RF-transparent (GPS works through the wall), Hall-sensor magnet flips through the lid, Qi-charge through the wall — same tricks as the Hammond and the RBF63 line. · enclosure · firmware: open-source

<https://www.digikey.com/en/products/detail/serpac/RBF53P06C10C/17100309>

CHF 21.97

Solderless alternative — bring-up only

If you don't want to touch the iron yet (or you want to verify the wiring before committing to a permanent build), the BOM anchors a solderless 4-SKU assembly on one distributor-tracked Part (`samtec-ffsd-07-100mm`) that lets you plug into JP2 without soldering. **Caveat:** a vibrating pumpfoil board can wiggle the cable loose mid-session — use this for bring-up / development, switch to the soldered rig below for production.

Reference videos (watch first)

- [ControllersTech — GPS Module and STM32 \(NEO-6M\)](#): generic STM32 + u-blox GPS over UART. NEO-6M but the wiring is identical to MAX-M10S (TX↔RX crossover, GND, 3.3 V). Watch this one if you watch one.
- [SparkFun — MAX-M10S Product Showcase](#): the specific GPS breakout from this BOM (PRT-18037). Shows the U.FL connector and the UART header pins you'll connect to.
- [Chris Park — STM32 + u-blox UART NMEA, wiring from 11:33](#): the closest-fit hookup demo. Uses M8N (same pinout as the MAX-M10S), shows 4 DuPont female leads plugged onto VCC/GND/TX/RX on the GPS module. Translate directly to the STEVAL: JP4 → 3 V & GND, JP2 → TX/RX (crossover: STEVAL TX → GPS RX, STEVAL RX → GPS TX).

There is no exact "STEVAL-MKBOXPRO + solderless JTAG cable + MAX-M10S" video — that combination is bespoke to this build. The two videos above cover the load-bearing concepts (UART wiring + the specific GPS module).

Shopping list (4 SKUs, ~CHF 17 total)

#	Part	Where / Price
1	Samtec FFSD-07-D-04.00-01-N — 14-pin 1.27 mm IDC ribbon, 100 mm (<i>BOM card above</i>)	DigiKey CH · ~CHF 9
2	1.27 → 2.54 mm SWD adapter PCB — the only piece without a Western-brand SKU (survey of DigiKey + Mouser + Farnell + Distrelec + ChipDepot + Bastelgarage + Amazon DE all came back empty)	AliExpress · ~CHF 3 · search "Cortex SWD 14-pin 1.27 to 2.54 adapter"
3	SparkFun PRT-12796 — F/F DuPont jumper wires, 6 in (20-pack; you need 4) (<i>BOM card above</i>)	DigiKey CH · ~CHF 4
4	Samtec SSW-107-01-G-S — 1×7 female header strip, 0.1" pitch (slips onto JP4 for the 3.3 V tap) (<i>BOM card above</i>)	DigiKey CH · ~CHF 1

Why this isn't a single SKU: a 14-pin 1.27 mm shrouded socket → loose 2.54 mm female DuPont cable is exactly what the buyer wants in one piece, but the category isn't stocked by any Western distributor (DigiKey, Mouser, Farnell, Distrelec, ChipDepot, Bastelgarage all checked); only generic Chinese OEM listings on Amazon/AliExpress fill it (~CHF 5–8, seller volatile). If you want to skip the adapter PCB + DuPonts, an AliExpress search for "14 Pin 1.27mm SWD to Dupont Cable" surfaces the single-cable version — but the BOM doesn't track it because no listing stays stable long enough.

Step 1 · JP4 power tap (3.3 V)

1. The JP4 connector is the 7-pin 2.54 mm strip on the side of the board (Multicomp MC-SVT1-S07-G).
2. **Set the JP4 domain switch to 3 V** (not 1.8 V).
3. Power on the box and **meter the JP4 3 V pin**: must read **3.25–3.40 V**. Power off again.
4. Slip a standard **2.54 mm 7-pin female header strip** onto JP4 (no solder — slides on). DuPont jumper from the 3 V pin → GPS VCC.

Step 2 · JP2 data lines (UART4)

1. Plug the JTAG cable into JP2 (FTSH-107 keyed/shrouded socket — only goes on one way).
2. Wire 3 DuPont jumpers from the adapter PCB's 2.54 mm side to the **SparkFun MAX-M10S breakout's UART header**:

STEVAL JP2 pin	Net (MCU)	→	SparkFun GPS pin
13	UART4_RX / PA1	→	TX
14	UART4_TX / PA0	→	RX
7 or 11	GND	→	GND

Note the TX/RX crossover (each end's "TX" pairs with the other's "RX"). The SparkFun breakout's UART header has male 2.54 mm pins populated — female DuPont leads slide on directly. **Do not touch SWD pins (4 / 6 / 8 / 10 / 12)** — they're the debug bus; bridging one means bricked debug access.

Step 3 · Bring-up test

1. Insert the SD card. Take the box **outdoors, clear sky**. Power on.
2. Cold fix is typically reached in **~30–60 s**. Confirm with a growing `GpsNNN.csv` on the SD card, or the BLE SensorStream `gps_valid` flag, or the antenna-test buzzer build.
3. No fix in a few minutes? **Swap pin 13 ↔ pin 14** — reversed TX/RX is by far the most common wiring mistake and swapping is harmless. Then re-check GND continuity and GPS VCC = 3.3 V under load.

When to switch to soldering: once bring-up works and you're ready for water trials. Vibration + spray + a loose cable will end a session early; the soldered rig below is the durable answer.

Soldering the u-blox MAX-M10S GPS to the STEVAL-MKBOXPRO

The MovementLogger firmware reads GPS over **UART4** on MCU pins **PA0 (TX)** / **PA1 (RX)** at **38400 baud, 8N1, 3.3 V logic**. The MAX-M10S is **not** on the box — it must be wired by hand to **JP2**, the 14-pin programming connector (Samtec FTSH-107-01-L-D, 1.27 mm pitch). Full source: [GPS_SOLDERING.md](https://github.com/movementlabs/steval-mkboxpro/blob/master/GPS_SOLDERING.md).

Tools you need

- Fine-conical-tip iron, flux, magnification (10× loupe or microscope), tweezers, steady bench.
- **30 AWG** enamelled / wire-wrap wire (≤ 0.5 mm), four lengths.
- Multimeter (continuity + DC volts).
- Kapton tape + a dab of hot glue for strain relief.

JP2 pinout — pin 1 = square pad / silk triangle

Pin	Net	Use?
1	1V8 rail (JTAG ref)	NOT power — 1.8 V, weak
4 / 6 / 8 / 10 / 12	SWDIO / SWCLK / SWO / JTDI / NRST	Leave alone (debug bus)
7	GND	OK for GPS GND
11	GND	OK for GPS GND
13	UART4_RX (PA1)	GPS TX → here
14	UART4_TX (PA0)	GPS RX → here

Pins 3, 5, 9 carry no signal — leave unconnected. **Meter-verify** pin 1 against the silkscreen triangle and continuity-check every pin before soldering — connector orientation differs between board builds.

Wiring — note the TX/RX crossover

GPS pin	→	JP2 pin	Net (MCU)
TX (GPS out)	→	13	UART4_RX (PA1)
RX (GPS in)	→	14	UART4_TX (PA0)
GND	→	7 or 11 (meter-verified)	GND
VCC 3.3 V	→	JP4 3 V pin (not JP2!)	Board 3 V rail

GPS power — 3.3 V from JP4

- JP4 is the 7-pin extension-board connector (Multicomp MC-SVT1-S07-G). Set its domain switch to **3 V** (not 1.8 V).
- Power the box and **meter the JP4 3 V pin** — must read **3.25–3.40 V** before connecting GPS VCC. Power down again before soldering VCC.
- GPS GND and box GND must be common — take GND from JP4 or a JP2 GND pin.
- No graceful shutdown: the Hall-sensor magnet cuts the whole supply rail at once (firmware F-PWR-5). The on-SD format tolerates it.

Safety — read before the iron is hot

- **3.3 V logic only**. PA0/PA1 are **not 5 V tolerant**. A 5 V GPS breakout's TX into PA1 will destroy the MCU — level-shift or use a 3.3 V-native MAX-M10S carrier (e.g. the SparkFun breakout above).
- **Never bridge to the SWD pins** (4/6/8/10/12) — solder bridge → bricked debug access.
- Keep each pad heated < 2 s. FTSH-107 pads lift if cooked.
- **Strain-relieve all four wires** with Kapton + hot glue close to JP2 — a tugged wire rips the pad off the PCB.

Procedure

1. **Power down completely** — switch S1 off, USB unplugged, LiPo disconnected.
2. **Locate JP2** (14-pin fine-pitch header by the MCU). Identify pin 1.
3. **Continuity-check before soldering** (meter, board off): JP2 pin 13 ↔ MCU PA1, JP2 pin 14 ↔ MCU PA0, candidate GND pin ↔ battery (–). Confirm, don't assume.
4. **Pre-tin** the three JP2 pads and the wire ends with flux.
5. **Solder the data wires**: pin 13 ← GPS TX, pin 14 ← GPS RX. Then **GND**.
6. **GPS VCC last**: with the board powered, confirm 3.3 V at the JP4 3 V pin, power down again, solder the VCC wire.
7. **Inspect under magnification** for bridges — especially pin 14 to pin 12 and onto the SWD pins.
8. **Strain-relieve** all four wires (Kapton + hot glue).

Bring-up test

1. Insert the SD card, take the box **outdoors with clear sky**, power on.
2. The firmware sends the u-blox UBX config on boot, then expects 10 Hz NMEA. First fix is typically reached within **~30–60 s cold**.
3. Confirm a fix by **any** of: a growing GpsNNN.csv on the SD card · the BLE SensorStream gps_valid flag going set · (antenna-test build, make GPS_ANTENNA_BEEP=1) the buzzer beeping N times every 10 s with N = satellites.

No fix after several minutes outdoors? — in order of likelihood

1. **Swap the two data wires** (pin 13 ↔ pin 14). TX/RX reversed is the single most common wiring mistake and swapping is harmless to try.
2. Re-check **GND** continuity (cold/again) and that GPS VCC really is 3.3 V *at the module* under load.
3. Confirm the module is a 3.3 V part and actually powered (most MAX-M10S carriers have a power LED).
4. Give it a longer cold-start with an unobstructed sky view; indoors it will never fix.